Phys. 2110 HW # 04 Solution

HW # 04 4.28, 38, 45, 49, 54, 61

4.28 (a) If the table is frictionless there is a net horizontal force on the combined of the two blocks, and block B accelerates in the direction of the pull. The friction force that B exerts on A is to the right, to try to prevent A from slipping relative to B as B accelerates to the right.
The free-body diagram is sketched below (a).
$\vec{f}$ is the friction force that B exerts on A and $\vec{n}$ is the normal force that B exerts on A.
(b) The pull and the friction force exerted on B by the table cancel and the net force on the system of two blocks is zero. The blocks move with the same constant speed and B exerts no friction force on A.
The free-body diagram is sketched below (b).

(a) (b)

4.38 $V_x^2 = V_{0x}^2 + 2a_x(x - x_0)$, if the reef weren't there the ship would stop in a distance

$$x - x_0 = -\frac{V_{0x}^2}{2a_x} = \frac{V_0^2}{2(F/m)} = \frac{mV_0^2}{2F} = \frac{(3.6\times10^7\,kg)(1.5\,m/s)^2}{2\times(8.0\times10^4\,N)} = 506\,m$$

so the ship would hit the reef.
The speed when the tanker hits the reef is found from
$V_x^2 = V_{0x}^2 + 2a_x(x - x_0)$

$$V = \sqrt{V_0^2 - 2\frac{Fx}{m}} = \sqrt{(1.5\,m/s)^2 - \frac{2\times(8.0\times10^4\,N)\times(500\,m)}{3.6\times10^7\,kg}} = 0.17\,m/s$$

and the oil should be safe.

4.45 (a) $x = (9.0\times10^3\,m/s^2)\,t^2 - (8.0\times10^4\,m/s^3)\,t^3$

$x = 0$ at $t = 0$

$t = 0.025\,s$, $x = (9.0\times10^3\,m/s^2)\times(0.025\,s)^2 - (8.0\times10^4\,m/s^3)\times(0.025\,s)^3 = \boxed{4.4\,m}$

So the length of the barrel must be 4.4m.

(b) $V_x = \frac{dx}{dt} = (1.8\times10^4\,m/s^2)\,t - (2.4\times10^4\,m/s^3)\,t^2$

At $t = 0.025\,s$, when the object reaches the end of the barrel.

$V_x = (1.8\times10^4\,m/s^2)\times(0.025\,s) - (2.4\times10^4\,m/s^3)\times(0.025\,s)^2 = \boxed{300\,m/s}$

(c) net force $\Sigma F_x = ma_x$

$a_x = \frac{dV_x}{dt} = 1.8\times10^4\,m/s^2 - (48.0\times10^4\,m/s^3)\,t$

(i) At $t=0$, $a_x = 1.8\times10^4\,m/s^2$, $\Sigma F_x = (1.50\,kg)\times(1.8\times10^4\,m/s^2) = \boxed{2.7\times10^4\,N}$

(ii) At $t = 0.025\,s$, $a_x = 1.8\times10^4\,m/s^2 - (48.0\times10^4\,m/s^3)(0.025\,s) = 6.0\times10^3\,m/s^2$

so $\Sigma F_x = (1.50\,kg)\times(6.0\times10^3\,m/s^2) = \boxed{9.0\times10^3\,N}$

4.49 The free-body diagram

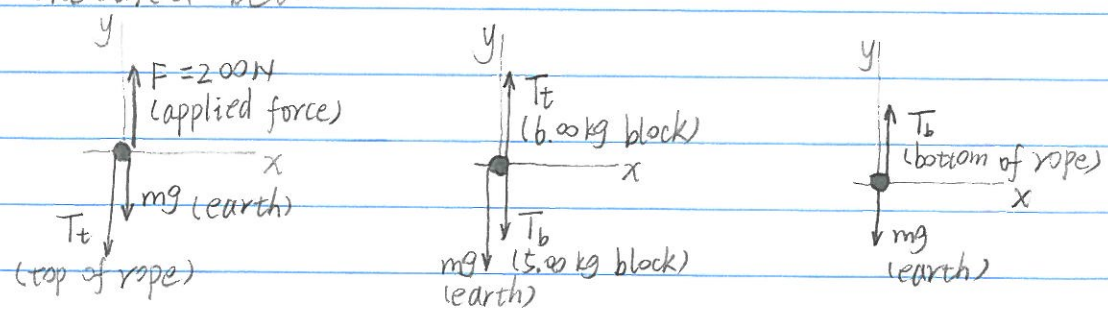
(a) If the gymnast climbs at a constant rate, there is no net force on the gymnast, so the tension must equal the weight, $T = mg$;

(b) No motion is no acceleration, so again $T = mg$;

(c) $T - W = T - mg = ma = m|\vec{a}|$ (the acceleration is upward, the same direction as the tension), so $T = m(g + |\vec{a}|)$;

(d) $T - W = T - mg = ma = -m|\vec{a}|$ (the acceleration is downward, the opposite direction to the tension), so $T = m(g - |\vec{a}|)$

4.54 (a) The free-body diagrams for each block and for the rope are sketched below



(b) Treat the rope and the two blocks together as a single object, with mass $m = 6.00\,kg + 4.00\,kg + 5.00\,kg = 15.0\,kg$

Take +y upward, the acceleration is upward.

$\Sigma F_y = ma_y \qquad F - mg = ma \qquad a = \frac{F - mg}{m} = \frac{200N - 15.0kg \times 9.80m/s^2}{15.0\,kg} = \boxed{3.53\,m/s^2}$

(c) Consider the force on the top block ($m = 6.00\,kg$), since the tension at the top of the rope (T_t) will be one of these forces.

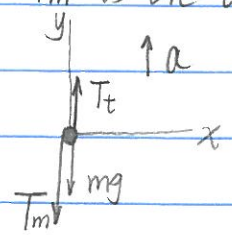
$$F - mg - T_t = ma$$

$$T_t = F - m(g+a) = 200N - 6.00\,kg \times (9.80\,m/s^2 + 3.53\,m/s^2)$$

$$= \boxed{120\ N}$$

(d) Consider the forces on the top half of the rope ($m = 2.00\,kg$).

T_m is the tension at the midpoint of the rope.



$$\Sigma F_y = ma_y$$

$$T_t - T_m - mg = ma$$

$$T_m = T_t - m(g+a)$$

$$= 120N - 2.00\,kg \times (9.80\,m/s^2 + 3.53\,m/s^2) = \boxed{93.3\,N}$$

4.61 (a) The equation of motion, $-cv^2 = m\frac{dv}{dt}$

that is $-\frac{c}{m}dt = \frac{dv}{v^2}$ and then integration

$$-\frac{ct}{m} = \frac{-1}{v} + \frac{1}{v_0}$$

where v_0 is the constant integration that gives $v = v_0$ at $t = 0$.

This expression can be rewritten as $v = \frac{dx}{dt} = \left(\frac{1}{v_0} + \frac{ct}{m}\right)^{-1}$

which may be integrated to obtain $x - x_0 = \frac{m}{c}\ln\left[1 + \frac{ctv_0}{m}\right]$

To obtain x as a function of v, the time t must be eliminated from the expression obtained after the first integration $\frac{ctv_0}{m} = \frac{v_0}{v} - 1$, so $x - x_0 = \frac{m}{c}\ln\left(\frac{v_0}{v}\right)$.

(b) Applying the chain rule, $\Sigma F = m\frac{dv}{dt} = mv\frac{dv}{dx}$

$-cv^2 = \left(v\frac{dv}{dx}\right)m$, $-\frac{c}{m}dx = \frac{dv}{v}$, integrating gives $x - x_0 = \frac{m}{c}\ln\left(\frac{v_0}{v}\right)$